Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building **backend services**, **SQL-backed data workflows**, and cloud-based applications across **Python, Java, SQL**, and data-centric platforms, with strong alignment to **data quality**, **reliability**, and scalable internal systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed a **multi-tenant data platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema design, indexing, and query optimization.
- Built **FastAPI** integration workflows across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation checks, reconciliation logic, and **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** platform serving **150+ instructors**, handling **10K+ submissions** with modular backend services, dashboard features, and structured data flows.
- Maintained **AWS RDS**-backed APIs and concurrent processing pipelines with **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and maintenance flows across **3 operating systems** with **20+ unit/integration suites**.
- Analyzed and improved API and ETL logic using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and structured business workflows, supporting **8+ modules** and **2M+ synchronized records**.
- Modeled users, resources, invoices, and notifications in **Azure SQL** while using Jenkins-driven testing and documentation to improve release reliability and reduce wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **Flask**, React, TypeScript, and Firebase, supporting **50K+ booking records** with sub-**200ms** APIs and real-time data synchronization.
- Designed booking, coupon, and payment workflows across client-facing and backend systems, improving booking confirmation speed by **40%** and reducing manual refund handling by **70%**.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices, Debugging
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL
Cloud & Data Systems: AWS (RDS, S3), Microsoft Azure, GCP, Docker, Kubernetes, Batch Processing, Simple Pipelines
Engineering Practices: Code Reviews, Documentation, Unit & Integration Testing, Data Quality, Reliability, Technical Design, Agile
AI/LLM Awareness: LLM Systems, GenAI Curiosity, Feature Engineering, Data-oriented Applications

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built a cloud-based document platform over **50K+ documents** with semantic retrieval, collaborative workflows, and **GCP**-backed indexing pipelines that reduced review time by **60%**.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, combining Java + TypeScript services, structured query-management workflows, and schema exploration from a unified interface.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a cloud-native fintech platform supporting **50K+ users** and **1M+ transactions** with containerized services, API-driven workflows, and production-grade deployment practices.